

CAMPUSCONNECT – PROBLEM STATEMENT

Lost & Found Management System for University Campuses

SLIIT SE2020: Web and Mobile Technologies | Year 2 Semester 2 — 2026

1. BACKGROUND & PROBLEM DEFINITION

In large higher education institutions like SLIIT, personal possessions such as student ID cards, smartphones, laptop power supplies, thumb drives, calculators, and wallets are lost every semester. The traditional campus approach relies on physical security notices, logbooks, or WhatsApp groups.

This creates significant operational drawbacks:

- **Low Recovery Rates:** Dispersed reports prevent owners from locating recovered belongings.
- **Verification Vulnerability:** Finders cannot reliably verify legitimate ownership before handover.
- **Heavy Administrative Overhead:** Campus security guards spend substantial time maintaining logs.
- **Lack of Transparency:** Students have zero visibility into the resolution status of their lost reports.

2. PROPOSED SOLUTION: CAMPUSCONNECT

CampusConnect is a small, complete, understandable mobile Lost & Found platform designed to resolve these challenges through structured relational workflows:

- **Centralized Mobile Repository:** Finders immediately publish found items with photos and campus locations.
- **Ownership Verification Engine:** Students submit claims containing secret or distinguishing proof.
- **Administrator Control:** Campus authorities evaluate proof and approve or reject claims.
- **Real-Time Status Tracking:** Statuses update dynamically across Active, Pending, Approved, and Claimed.

3. CORE OBJECTIVES

1. Deliver a responsive cross-platform mobile frontend using React Native, Hooks, and React Navigation.
2. Establish a secure Node.js & Express.js REST API with complete CRUD on primary & related entities.
3. Enforce relational modeling between User, Item, and Claim in MongoDB Atlas using Mongoose references.
4. Execute all 7 domain business logic rules strictly at the API controller layer.
5. Implement image uploads via Multer with validation and static cloud asset serving.

4. SYSTEM SCOPE & USERS

- **Target Users:** SLIIT Undergraduate Students, Faculty, and Campus Security/Administration.
- **Boundaries:** Strictly focused on lost and found management, ownership claims, and role authorization.
- **Excluded (Per Assignment Scope):** Payment gateways, chats, web admin dashboards, or social feeds.